Heterogeneous Effects of State Enterprise Zone Programs in the Shorter Run and Longer Run

Economic Development Quarterly
2021, Vol. 35(2) 91–107

DOI: 10.1177/0891242420982341
journals.sagepub.com/home/edq
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Abstract

The authors take up two questions that have not been explored in research on enterprise zones. First, does a considerably longer-run perspective on the effects of state enterprise zones lead to different answers? And second, are there heterogeneous effects of enterprise zones that depend on the set of incentives these programs offer, which can vary widely? The results indicate that whether state enterprise zone programs were observed through a longer-term lens, or through the lens of program heterogeneity, the authors generally did not find any consistent indication of beneficial effects of state enterprise zone programs; if anything, the longer-run effects are negative. The lack of positive effects is consistent with most of the prior evidence that focuses on effects that are short term and homogeneous.

Keywords

enterprise zones, longer-term effects, program heterogeneity, unemployment, poverty

The body of evidence on place-based policies, particularly enterprise zones, provides little reason to think these policies are effective. As reviewed recently in Neumark and Simpson (2015), most research on enterprise zone programs failed to find strong evidence that they contribute to job growth or reduce poverty.

There are some exceptions. Busso et al. (2013) found strong evidence that some federal enterprise zones (Empowerment Zones) generated strong job growth; although even then, Reynolds and Rohlin (2015) reported evidence suggesting that these zones may have this effect through displacing the most disadvantaged original residents of these zones to other areas.

Also in contrast to most of the literature, Ham et al. (2011) found that both state and federal enterprise zones established in the 1990s significantly improved labor market outcomes, with sharp reductions in poverty from state enterprise zones and sharp reductions in unemployment and poverty, and sharp increases in income measures, from federal enterprise zones (Empowerment Zones and Enterprise Communities).[1]

In this article, we take up two questions that have not been explored in research on enterprise zones. First, does a considerably longer-run perspective on the effects of state enterprise zones lead to different answers? It is possible that a longer-run perspective will provide more evidence of beneficial effects than the shorter-run evidence, especially if enterprise zones help "jump start" economic activity in targeted areas, creating some self-sustaining gains. For example, one rationale for enterprise zones specifically, and place-based policies more generally, is that they may generate benefits from agglomeration externalities intended to lead to persistent gains in economic activity from a shorter-term policy intervention (see Kline & Moretti, 2014; Neumark & Simpson, 2015). On the other hand, it is possible that any beneficial effects of enterprise zones, if they exist, would be muted in the longer run. Over time, enterprise zone incentives could be capitalized into immobile production factors, perhaps especially land (e.g., Burnes, 2012; Landers, 2006), diminishing the cost advantage of the zones for businesses. Zones may also become less effective over time as additional zones are created,[2] or if positive effects spill over to other areas. Of course, in this latter case the short-run evidence would provide the greatest hope of uncovering beneficial effects of enterprise zones, which it generally does not.

Most research on the effects of enterprise zones takes a relatively short-run perspective. For example, Busso et al. (2013) and Reynolds and Rohlin (2015) estimated the impacts of federal enterprise zones created in the mid-1990s on changes in census tracts over the 1990s (for their main analysis). Ham et al. (2011) pursued the same strategy, looking at both federal and state enterprise zones. Elvery (2009)

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used a propensity score method—rather than difference-in-differences methods—to measure the impact of enterprise zones created in California and Florida between 1986 and 1990 on outcomes in 1990 data. Billings (2009) considered a mix of shorter-run and longer-run effects by using establishment-level data averaged between 1990 and 2000 to study zones created in the late 1980s.

There are some exceptions that take a longer-run approach. Neumark and Kolko (2010) compared short-run and longer-run effects of California's enterprise zone program; however, these different effects are captured only as shifts in employment levels versus growth rates, which is quite restrictive. Freedman (2013) studied effects of Texas enterprise zones over a window that can be up to 6 years after an enterprise zone is established, but the study simply focuses on annual employment growth instead of breaking out longer-term effects. There are also a few earlier studies of employment effects with somewhat longer-run perspectives, using research designs that are less compelling (see Neumark & Simpson, 2015). Boarnet and Bogart (1996) on enterprise zones in New Jersey, Papke (1994) in Indiana, and O'Keefe (2004) in California estimated the effects of enterprise zones on employment growth up to 6 years after their creation; these studies present conflicting evidence. Our analysis takes a considerably longer-run perspective and focuses on state enterprise zones (and is done both with and without incorporating program heterogeneity).

The second question we consider is whether there are heterogeneous effects of enterprise zones that depend on the set of incentives these programs offer, which can vary widely. Evidence on this question could, in principle, help policy makers design more effective enterprise zone programs. Although the existing research on enterprise zones is generally discouraging, there is a strong theoretical expectation that well-designed hiring credits could boost job creation. Estimating the effects of different features of state enterprise zone programs could identify some features of these programs that are more effective in job creation or other goals, such as poverty reduction.

Moreover, there is a smattering of evidence pointing to some heterogeneity in the job creation impacts of enterprise zones, suggesting that certain program designs and/or improved targeting could produce positive impacts. For example, Neumark and Simpson (2015) suggested that the initial federal Empowerment Zones studied by Busso et al. (2013) may have been more effective at creating jobs than subsequent versions of the policy (or other types of enterprise zones), because the initial Empowerment Zones included large block grants, which may have been spent on social services or other investments that boosted employment. In addition, Kolko and Neumark (2010) incorporated survey data from enterprise zone administrators into an analysis of the effects of California's state enterprise zones and found that a subset of efforts (e.g., marketing zone incentives) was associated with job gains. Finally, although one step removed from enterprise zones, recent work on general statewide hiring credits indicated that such credits are much more likely to lead to job creation when governments can "claw back" the credits if jobs are not created (Neumark & Grijalva, 2017).

Given our emphasis on enterprise zone program heterogeneity, in this article we focus on state enterprise zone programs, which differ along many dimensions. In doing so, we build on the work by Ham et al. (2011), who estimated the effects of enterprise zones in 13 states—albeit, in contrast to our article, without considering heterogeneity across state enterprise zone programs.

We estimate effects on outcomes related to wages and employment. The predicted positive effects on wages and employment from place-based policies like enterprise zones stem from a number of sources, including agglomeration, mitigating spatial mismatch, network effects on hiring (especially when local hiring is incentivized), and the direct impact of reductions in labor costs and other costs from tax credits or incentives. In particular, we analyze effects on the following five outcomes: the unemployment rate; the poverty rate (at the person level); the fraction of households with wage and salary income; average wage and salary income (measured at the household level in 2000 dollars and including zeros); and the level of employment. These outcomes and the ways we define them follow Ham et al. (2011). More broadly, though, they parallel much of the enterprise zone literature that assesses enterprise zones based on their promised effects on employment and wages, which would in turn be predicted to positively affect the other outcomes we study. The exception is that some studies focus instead (or as well) on establishment counts or births and deaths. These are not available in the data we use in this article.[3]

Data

There are two significant data challenges in this research. The first is to code the heterogeneity in state enterprise zone policies and the second is to map, over a longer period, the census tracts treated and not treated by state enterprise zone programs. We discuss these in turn.

State Enterprise Zone Policy Heterogeneity

There is significant heterogeneity in enterprise zone benefits across states. We use multiple data sources to classify the heterogeneity of state enterprise zone benefits for the 13 states we examine in this article. Our primary source of information on enterprise zone benefits comes from reading each state's original legislative documents relating to the enactment and amendment of enterprise zone programs and benefits. State bills and statutes relating to enterprise zones were accessed through HeinOnline and LexisNexis. The potential features by which state enterprise zone programs could be classified are too numerous and idiosyncratic to be directly usable in an empirical analysis. Thus, the information in these documents was used to classify all enterprise zone benefits for all 13 states into four broad categories. We summarize the details of each state's enterprise zone benefits in supplementary Appendix A, available online. The four categories of benefits we use in our analysis are listed in Panels 1 to 4 of Appendix Table A1, available online; we discuss these categories below. We also classify states by whether each state's enterprise zone legislation requires employees to be zone residents for a business to be eligible for enterprise zone incentives (Panel 5 of Appendix Table A1, available online), and whether these "live-in" requirements are specific to hiring credits (Appendix Table A2, available online).

Our classification process is exhaustive in that every benefit offered by each of the 13 states between 1990 and 1999 is classified as belonging to one of the broad categories we use. To verify that the benefits described in the letter of the law match the incentives provided and marketed in practice to businesses (and in some cases employees), each state's list of benefits was cross-referenced with peer-reviewed journal articles,[4] state tax forms,[5] state agency reports,[6] and other reports[7] about enterprise zones. See online Appendix B for full details on the enterprise zone benefits provided by each of the 13 states.[8]

The four broad categories into which we classify enterprise zone benefits are:

- Financing
- Tax credits for investments
- Miscellaneous
- Hiring incentives

Financing options are provided by six states and range from interest-free loans (Florida) to providing tax deductions for contributions to a corporation designated for providing loans to businesses operating in enterprise zones. The provision of financing options may facilitate new firm entry into enterprise zones on the extensive margin while enabling expansion of existing firms on the intensive margin.

All state enterprise zone programs provide tax credits for investments—the second category. The definition of qualified investments varies across states from very vague (Nebraska) to rather specific (Ohio's environmental remediation tax credit). Generally, states provide incentives for investments in property, such as real estate, construction materials, and any other expansion or rehabilitation.

We also capture a miscellaneous category of enterprise zone benefits that do not fit easily into a broad category. This category includes benefits provided optionally by local municipalities, day care reimbursements, loss limitations, and the sale or lease of public property for business use. Among these, our analysis focuses on the benefits that come in the way of local property tax incentives.[9]

Finally, our fourth category is hiring credits or hiring incentives, which are common to enterprise zones and attract the most attention in the literature. Hiring incentives generally reduce the wage bill for existing employees through tax credits, provide a flat credit for newly hired employees, or provide exemptions for payroll and unemployment insurance taxes. These incentives are designed to encourage firms operating in enterprise zones to expand hiring or to hire new employees. Therefore, on the extensive margin, hiring incentives are likely to increase employment and to decrease unemployment.

As online Appendix B shows, our extensive research on enterprise zone legislation captures the intensity of the program benefits. However, we do not incorporate information on the intensity of incentives into our main analysis because this is very difficult to do in a systematic, consistent, and tractable manner. For example, looking at the first two states, the hiring credit information for California includes the magnitude of the hiring credit and the financing incentive for Colorado captures the tax parameters. However, such information is not always provided, and even when it is, not in a way that makes things directly comparable. (As examples, in Rhode Island the credit varies with whether the hire is a zone resident and in Hawaii with the percentage of employees who are low income.) There is thus simply no straightforward way to assign intensity measures to the credits. Our hope, nonetheless, is that other researchers might find ways to use the information captured in the online Appendix B to learn more about the effects of different enterprise zone features, including the strength of incentives.[10]

Construction of Longer-Run Enterprise Zone Data Set

The second data challenge for this project was to create a long-term classification of census tracts (our unit of observation) by enterprise zone status. Our starting point is the tracts identified as being initially designated as state enterprise zones tracts in the 1990s, using the data from Ham et al. (2011), which they provided.[11]

Some tracts that Ham et al. (2011) used as controls for their 1980 to 2000 analysis (also excluding federally designated tracts in the 1990s) are later designated as state or federal enterprise zones between 2000 and 2010. Including these tracts as controls in our longer-run analysis would attenuate our estimates. To identify the 2000 tracts that became state enterprise zones between 2000 and 2010, we did extensive work with agencies in each of the 13 states to generate geocoded maps of enterprise zone coverage, as best we could. We were able to identify these more recent state enterprise zone changes for all of Ham et al.'s 13 states except for Massachusetts and Wisconsin. We detail our efforts in online Appendix C.[12]

For federal enterprise zones, Ham et al. (2011) identified the tracts designated as federal enterprise zones in the mid-1990s. However, two additional waves of federal enterprise zones occurred in 1999 and 2001 that are not identified in their data. We identify all tracts enacted as federal enterprise zones in any of the three waves, which extend through 2001, using data from Busso et al. (2013).

In our work with state-level agencies to identify the tracts designated as state enterprise zones in 2000 to 2010, we identified—for three states—information on tracts designated in the prior decade (the 1990s). For these three states, we identified some tract designations that conflicted with the designations in the 1990s identified in the Ham et al. (2011) data. Moreover, Ham et al. provided no documentation of the underlying source material for their tract designations, such as original maps or underlying sources that they used to determine treatment status.[13] Thus, for the three states where we have more reliable information—California, Illinois, and Ohio—we use the designations we obtained from the state-level agencies, rather than the Ham et al. coding.[14,15] We impose the same sample restrictions as Ham et al. by dropping partially treated tracts where fewer than 50% of the tract's area is covered by an enterprise zone, dropping tracts designated as first-round federal enterprise zones, and dropping any tracts enacted as enterprise zones during the 1980s.[16]

Data on Outcomes by Tract

We use the NCDB to measure outcomes in treated and control tracts for the period 1980 to 2010. Census tract geography changes over time, but the NCDB provides consistent tract definitions over time.[17] We make year 2000 tracts our preferred geography because this directly maps into the enterprise zone geography used in Ham et al. (2011). The 1980 to 2000 NCDB is available in terms of 2000 tracts, but the 2010 NCDB, which we need to study longer term effects of enterprise zones established in the 1990s, is only available in 2010 tract geography. We utilize a program called Backwards LTDB, provided through the Longitudinal Tract Data Base, to map the 2010 NCDB data into 2000 tracts.[18]

We make several sample restrictions to the NCDB data. First, we drop tracts with zero population counts in 1980. This occurs, according to our best understanding, because of redrawn tract boundaries in terms of 2000 geography. It turns out that Illinois had relatively few tracts designated as enterprise zones and all of them are dropped because of this restriction; thus, Illinois is not included in our empirical analysis. Second, we only keep data for the 13 states in Ham et al. (2011)'s analysis. Third, we remove one tract that appears in the 1980 to 2000 NCDB but does not appear in the 2010 NCDB after processing by the Backwards LTDB program. Fourth, we drop tracts that have missing values for any of the five outcomes in any year of the NCDB. We do this to ensure that our estimation is always done for the same sample, regardless of the outcome.

In all cases, we study effects on the five outcomes that Ham et al. (2011) study: the unemployment rate, the poverty rate, the fraction of households with wage and salary income, average wage and salary income, and the level of employment.

Analysis Samples

In most cases, we report results for three different samples. We first use all possible data. We then drop the data from Massachusetts and Wisconsin because, as explained above, we were unable to obtain information on changes in the enterprise zone designation of tracts in these states during the 2000s. Hence, we are unsure of whether some tracts in these states became designated as enterprise zones. Finally, we also limit the set of control tracts, excluding those that were later designated as state enterprise zones in the 2000s (or as federal zones in 1999 to 2001). This is our cleanest estimation because it avoids contaminated controls, so we emphasize results for this sample but also show some results for the other samples. Table 1 shows the numbers of control and treated tracts by state, as well as information on the tracts that are contaminated controls, and those for which we do not have information for the 2000s (i.e., Massachusetts and Wisconsin).

Methods and Results

Our analysis proceeds in several steps, and we explain these sequentially, after first describing the methods we use.

Empirical Approach

Our estimation framework, at its core, follows the broad strategy used by Ham et al. (2011), albeit with important differences that—the evidence indicates—give more reliable results. We begin with a shorter-run analysis that parallels their analysis, then turn to our longer-run analyses, and the analysis of program heterogeneity in both the shorter and longer run. But the key identification strategy can be explained in the context of our initial, shorter-run analysis.

Ham et al. (2011) study zones were created between 1990 and 2000. They computed the difference in outcomes between 1990 and 2000 for tracts where zones were created, which we denote $\Delta Y_{cs1}{}^{T}$, where Y is the dependent variable of interest, the "c" subscript denotes census tracts, the "s" subscript denotes states, the 1 subscript denotes that the difference is for the posttreatment period, and the "T" superscript denotes that this difference is computed for treated tracts. They subtract from this the pretreatment difference—between 1980 and 1990—for the same tracts, $\Delta Y_{cs0}{}^{T}$.

Two of the sets of control tracts that Ham et al. (2011) used are the nearest tract to each treated tract and the average

Table 1. Treated and Control Tracts for Alternative Analysis Samples.

	Enterprise zone in 2000: NCDB			Enterprise zone in 2000: limited controls				
	Control	Treated	Total	Control	Treated	Contaminated controls (either fed or state enterprise zone)	2000-2010 enterprise zone data not available	Total
California	6,004	325	6,329	5,243	325	761	0	6,329
Colorado	823	14	837	551	14	272	0	837
Florida	2,753	67	2,820	2,380	67	373	0	2,820
Hawaii	255	9	264	198	9	57	0	264
Illinois	2,270	0	2,270	2,077	0	193	0	2,270
Massachusetts	699	567	1,266	0	0	0	1,266	1,266
Nebraska	219	3	222	210	3	9	0	222
New York	4,158	115	4,273	3,476	115	682	0	4,273
Ohio	1,155	981	2,136	510	981	645	0	2,136
Oregon	478	48	526	401	48	77	0	526
Rhode Island	194	27	221	156	27	38	0	221
Virginia	1,179	29	1,208	978	29	201	0	1,208
Wisconsin	698	25	723	0	0	0	723	723
Total	20,885	2,210	23,095	16,180	1,618	3,308	1,989	23,095

Note. NCDB = Neighborhood Change Database.
Source. Authors' computations.

over all contiguous tracts.[19] In either case, there is a control "tract" matched to each treated tract—a single tract in the case of the nearest tract controls, and an average tract in the case of the contiguous tract controls (averaging across the set of contiguous tracts); the nearest or contiguous tracts are in the same state as the treated tracts. In either approach, they construct a difference-in-differences for the control tract matched to each treated tract c, $\Delta Y_{cs1}{}^{C} - \Delta Y_{cs0}{}^{C}$. To estimate a common effect of state enterprise zones using these two types of controls, they form the triple difference (DDD) as the difference between the two double differences and estimate a simple regression of this DDD on an intercept (using random county effects), as in the equation:

$$\left\{\Delta Y_{cs1}{}^{T} - \Delta Y_{cs0}{}^{T}\right\} - \left\{\Delta Y_{cs1}{}^{C} - \Delta Y_{cs0}{}^{C}\right\} = \beta + \varepsilon_{cs}. \tag{1}$$

This DDD estimator identifies the effect of enterprise zone designation from the change in the dependent variable in treated tracts from 1990 to 2000 minus the change from 1980 to 1990, relative to the same difference-in-differences in control tracts.

However, as highlighted in Neumark and Young (2019), simply using nearby or contiguous tracts as controls does nothing to ensure that there are common pretrends in the treated and control tracts.[20] And indeed, the 1980 to 1990 trends are often strongly different in the treated and control tracts; there is strong statistical evidence against the equality of these trends from 1980 to 1990 for both state and federal enterprise zones, and—even more strikingly—the trends are sometimes in opposite directions. The same analysis is also applied to what might be considered more natural controls—tracts that applied for enterprise zone status but were rejected. This type of analysis can be used in the context of federal Empowerment Zones and Enterprise Communities. While the deviations in prior trends were not as marked using this approach, the prior trends were sometimes significantly different.

To better match the pretreatment trends in outcomes between the treated and control tracts, we employ a data-driven approach to select control tracts, using propensity score methods.[21] Using this approach, we match treatment and control tracts to minimize the differences in pretreatment outcomes in terms of levels and changes for the outcomes we study.[22] For each program, we match a single control tract to each treated tract by matching on the 1980 and 1990 levels for all five outcomes (i.e., we match on a total of 10 covariates); matching on levels in both years effectively matches on changes as well.[23] Our estimator is based on a comparison of the double difference (the 1990 to 2000 change minus the 1980 to 1990 change) between each treated tract and its matched control tract; hence, we estimate the average treatment effect on the treated (ATT).[24,25]

A standard procedure to verify the effectiveness of the propensity score matching procedure is to compare pretreatment means between treated tracts and controls selected using the matching, with the most relevant means in our case being pretreatment changes in the outcomes. Neumark and Young (2019) have shown that the propensity score matching procedure does a good job of matching 1980 to 1990

Table 2. Shorter-Run Estimates of the Effect of Enterprise Zones (EZs) Using Propensity Score Matched Controls (1990-2000).

	Unemployment rate (%)	Poverty rate (%)	Fraction of households with wage and salary income (%)	Average household wage and salary income ($2000)	Employment
Panel 1: No limits on controls					
EZ	−1.003*** (0.243)	0.0121 (0.318)	0.565* (0.335)	−539.4* (322.4)	−20.39 (12.64)
N (EZ)	23,095 (2,210)	23,095 (2,210)	23,095 (2,210)	23,095 (2,210)	23,095 (2,210)
Panel 2: Excluding Massachusetts and Wisconsin					
EZ	−0.842*** (0.217)	−0.102 (0.424)	0.501* (0.289)	−606.6 (468.2)	−29.15* (16.01)
N (EZ)	21,106 (1,618)	21,106 (1,618)	21,106 (1,618)	21,106 (1,618)	21,106 (1,618)
Panel 3: Only limited controls					
EZ	−1.066*** (0.343)	0.400 (0.503)	1.009*** (0.383)	−1,445*** (444.9)	−24.83 (16.01)
N (EZ)	17,798 (1,618)	17,798 (1,618)	17,798 (1,618)	17,798 (1,618)	17,798 (1,618)

Note. Each cell is an estimate of the average shorter run (1990-2000) effect of enterprise zones from a separate propensity score matched probit model that matches treatment to control tracts using 10 covariates: 1980 and 1990 tract-level unemployment rate, poverty rate, fraction of households with wage and salary income, average wage and salary income, and employment. The estimates presented are average treatment effects on the treated. The control tracts are not restricted to the same state as the treated tracts.
Source. Authors' computations.
*$p < .1$. **$p < .05$. ***$p < .01$.

changes. For example, the prior change in the unemployment rate from 1980 to 1990 in enterprise zone tracts is 1.4 percentage points, compared with about 1.3 percentage points in the matched controls, resulting in a small and insignificant difference of −0.146. For the samples used in this article (described below), for each of our five outcomes, we never reject the equality of pretreatment parallel trends using propensity score matched controls, even at the 10% level (see online Appendix Table D1).

Our strategy for estimating longer run effects of state enterprise zones—while still treating these programs as homogeneous—is a simple extension of our shorter-run analysis. We maintain the same matched control tracts, since these are based on 1980 to 1990 changes, but redefine our outcome variable as the 1990 to 2010 (instead of to 2000) changes in outcomes.

For both the shorter-run and longer-run analyses, we estimate models imposing homogeneous effects across states. We then also estimate models with separate effects by state. These are independently interesting, but also a crucial input into our analysis of the effects of different enterprise zone benefits. For our analysis of program heterogeneity, we use a two-step estimation strategy. We take the state-specific estimates of the effects of enterprise zones, for both shorter-run and longer-run effects (first step), and then regress these estimates on dummy variables for the different enterprise zone benefit categories (second step). In the second step of the estimation, we weight by the number of treated tracts in that state. In a regression framework, one could instead estimate a single model with interactions between the enterprise zone dummy variable and dummy variables for enterprise zone benefit categories. However, this cannot be extended in a straightforward manner to the propensity score framework. We did, though, estimate this type of regression model using only the controls selected by the propensity score matching. This yields the same estimates, but different (and incorrect) standard errors, because this estimation does not consider the uncertainty associated with propensity scores being estimated instead of known.[26]

Shorter-Run Estimation of the Effects of State Enterprise Zones

We begin with estimation of the shorter-run effects of state enterprise zones. The results are reported in Table 2. The table has three panels: Panel 1 uses all possible data, Panel 2 drops the data from Massachusetts and Wisconsin, and Panel 3 excludes from the control tracts those that were later designated as state enterprise zones in the 2000s (or as federal zones in 1999 to 2001)—our cleanest estimation.

The evidence in Table 2 (which essentially replicates our earlier work) generally shows little consistent evidence of beneficial effects of enterprise zones on the outcomes we study. The two exceptions are a decline in the unemployment rate of about 1 percentage point, and a small increase in the fraction of households with wage and salary income (by 0.5 to 1 percentage point). For the other outcomes, the evidence is not generally in the direction of beneficial effects of enterprise zones; two of the point estimates for poverty are positive, and those for average wage and salary income and employment are all negative (and strongly significant in Panel 3). On the one hand, the evidence of an approximate 1 percentage point decline in the unemployment rate across the three panels of Table 2 points to some labor market gains. But the absence of evidence of an employment increase or poverty decline leads us to interpret any evidence of gains (even limited to jobs) as ambiguous at best.

Table 3. Shorter-Run State-Specific Estimates of the Effect of Enterprise Zones (EZs): Only Limited Controls (1990-2000).

	Unemployment rate (%)	Poverty rate (%)	Fraction of households with wage and salary income (%)	Average household wage and salary income ($2000)	Employment
EZ × California	−0.241 (0.275)	2.851** (1.142)	1.992* (1.117)	−752.2 (1,033)	−17.13 (44.87)
N (EZ)	16,505 (325)	16,505 (325)	16,505 (325)	16,505 (325)	16,505 (325)
EZ × Colorado	−2.285 (2.435)	−4.609 (6.694)	0.461 (6.070)	3,544 (4,864)	61.21*** (20.92)
N (EZ)	16,194 (14)	16,194 (14)	16,194 (14)	16,194 (14)	16,194 (14)
EZ × Florida	−2.869 (3.066)	0.0775 (5.492)	2.709 (3.284)	−411.3 (1,946)	−191.2 (153.8)
N (EZ)	16,247 (67)	16,247 (67)	16,247 (67)	16,247 (67)	16,247 (67)
EZ × Hawaii	1.128 (3.922)	−1.005 (8.171)	2.073 (7.612)	−183.0 (5,502)	312.8 (376.4)
N (EZ)	16,189 (9)	16,189 (9)	16,189 (9)	16,189 (9)	16,189 (9)
EZ × Nebraska	−4.568 (10.48)	−1.883 (11.47)	2.044 (16.52)	7,383 (11,300)	−453.3*** (125.4)
N (EZ)	16,183 (3)	16,183 (3)	16,183 (3)	16,183 (3)	16,183 (3)
EZ × New York	1.687** (0.747)	4.411*** (1.548)	−0.767 (1.604)	−4,351*** (1,350)	−165.5*** (57.27)
N (EZ)	16,295 (115)	16,295 (115)	16,295 (115)	16,295 (115)	16,295 (115)
EZ × Ohio	−2.094*** (0.353)	−1.997*** (0.437)	0.646* (0.345)	−1,882*** (597.2)	−37.91* (22.04)
N (EZ)	17,161 (981)	17,161 (981)	17,161 (981)	17,161 (981)	17,161 (981)
EZ × Oregon	−1.359 (1.565)	−3.492 (2.510)	−2.004 (4.320)	417.8 (2,081)	103.4 (79.78)
N (EZ)	16,228 (48)	16,228 (48)	16,228 (48)	16,228 (48)	16,228 (48)
EZ × Rhode Island	−0.235 (1.609)	3.934* (2.180)	−0.954 (3.641)	−9,734*** (2,154)	−216.7 (132.0)
N (EZ)	16,207 (27)	16,207 (27)	16,207 (27)	16,207 (27)	16,207 (27)
EZ × Virginia	0.870 (1.109)	1.463 (3.992)	−0.667 (4.248)	−2,035 (1,597)	−122.0 (136.7)
N (EZ)	16,209 (29)	16,209	16,209	16,209	16,209

Note. Each cell is an estimate of the average shorter-run (1990-2000) effect of enterprise zones from a separate propensity score matched probit model that estimates the effect of enterprise zones in one state using potential controls from all states. (The only exception is for Ohio, where we encountered convergence problems with probit estimation and used a logit model instead; the probit converged, but only with a slacker tolerance.) The propensity score matched model matches on 10 outcomes: 1980 and 1990 tract-level unemployment rate, poverty rate, fraction of households with wage and salary income, average wage and salary income, and employment. The estimates presented are average treatment effects on the treated. All estimates in this table exclude tracts in Massachusetts and Wisconsin, and all control tracts that were designated as either state or federal enterprise zones in the 2000s.
Source. Authors' computations.

We also estimated the models for the same five dependent variables by state. We show, in Table 3, the estimates corresponding to Panel 3 of Table 2—the most restrictive but cleanest sample.[27] There is a smattering of significant effects estimated in Table 3. This is more the case for some states than others. Furthermore, where there is statistically significant evidence, more of it points to evidence of harmful rather than beneficial effects. Examples of adverse effects include the effect on poverty in California; the effects on average income and employment in Ohio; the effects on poverty and average income in Rhode Island; and the effects on the unemployment rate, poverty, average income, and employment in New York.[28]

On the other hand, there is some statistically significant evidence in the opposite (i.e., beneficial) direction: in California, raising the fraction with wage and salary income; in Colorado, raising employment; and in Ohio, reducing the unemployment rate and poverty rate and increasing the fraction with wage and salary income.

This conflicting evidence by state, coupled with similar conflicting evidence on enterprise zone effects when estimating homogeneous effects for all states, reinforces the conclusion from most of the literature that it is difficult to find compelling evidence of beneficial effects of state enterprise zones. One obvious question posed by the differences in estimates across states, however, is whether the differences are accounted for by variation in state enterprise zone features. However, before considering evidence on this question, we first turn to longer-run estimates.

Longer-Run Estimation of the Effects of State Enterprise Zones

Turning to the longer-run analysis, the results imposing homogeneous effects across states are reported in Table 4, for all three samples, in Panels 1 to 3. The longer-run effects are uniformly negative, with every single estimate in Table 4 pointing to adverse effects—raising unemployment and poverty rates, and lowering the fraction with wage and salary income, average income, and employment. Also, many of these estimates are statistically significant. Thus, there is no evidence of positive effects of state enterprise zone programs in the longer run, and indeed the evidence points in the opposite direction.

Table 4. Longer-Run Estimates of the Effect of Enterprise Zones (EZs) Using Propensity Score Matched Controls (1990-2010).

	Unemployment rate (%)	Poverty rate (%)	Fraction of households with wage and salary income (%)	Average household wage and salary income ($2000)	Employment
Panel 1: No limits on controls					
EZ	0.850*** (0.269)	1.604*** (0.372)	−0.447 (0.364)	−3,758*** (682.1)	−123.7*** (21.67)
N (EZ)	23,095 (2,210)	23,095 (2,210)	23,095 (2,210)	23,095 (2,210)	23,095 (2,210)
Panel 2: Excluding Massachusetts and Wisconsin					
EZ	0.644** (0.306)	1.215** (0.494)	−0.632* (0.363)	−4,085*** (820.4)	−106.4*** (23.70)
N (EZ)	21,106 (1,618)	21,106 (1,618)	21,106 (1,618)	21,106 (1,618)	21,106 (1,618)
Panel 3: Only limited controls					
EZ	1.577*** (0.304)	2.490*** (0.538)	−0.323 (0.404)	−6,347*** (954.2)	−120.8*** (24.34)
N (EZ)	17,798 (1,618)	17,798 (1,618)	17,798 (1,618)	17,798 (1,618)	17,798 (1,618)
Panel 4: Only limited controls, dropping enterprise zone tracts de-designated in 2000s					
EZ	1.436*** (0.271)	2.303*** (0.576)	−1.314*** (0.465)	−5,099*** (968.9)	−181.2*** (28.44)
N (EZ)	17,696 (1,516)	17,696 (1,516)	17,696 (1,516)	17,696 (1,516)	17,696 (1,516)

Note. Each cell is an estimate of the average longer-run (1990-2010) effect of enterprise zones from a separate propensity score matched probit model that matches treatment to control tracts using 10 covariates: 1980 and 1990 tract-level unemployment rate, poverty rate, fraction of households with wage and salary income, average wage and salary income, and employment. The estimates presented are average treatment effects on the treated. The control tracts are not restricted to the same state as the treated tracts. In Panel 4, we assume that if we do not have data showing a tract continuing to be designated as an enterprise zone in the 2000s, then it was de-designated.
Source. Authors' computations.
$^{*}p < .1$. $^{**}p < .05$. $^{***}p < .01$.

Finally, Panel 4 reports estimates, based on the sample in Panel 3, where we drop from the treatment group enterprise zone tracts that we confirmed as "de-designated" as enterprise zone tracts in the 2000s.[29] There are two ways to think about this issue. One is that we might expect to find more positive longer-run effects when we drop from the longer-run analysis tracts where enterprise zone incentives were eliminated. Alternatively, we might be particularly interested in the difference in results for tracts that remained enterprise zones and tracts that were de-designated, perhaps because the strongest policy argument for enterprise zones is that they can change economic conditions in ways that create some self-sustaining benefits. In general, the results are little changed from Panel 3, with estimated adverse effects for all outcomes, and roughly similar magnitudes. Overall, then, there is no clear evidence that effects were very different in de-designated tracts.

Table 5 reports the longer-run estimates by state, paralleling Table 3. Again, we show the estimates corresponding to Panel 3 of Table 4.[30] Compared with the shorter-run estimates in Table 3, there are fewer significant estimates of longer-run effects. Furthermore, there are only two estimates in the table indicating a positive effect: raising average wage and salary income in Colorado and reducing the poverty rate in Oregon. In contrast, there are 11 statistically significant estimates pointing to a worsening of outcomes. These are mainly for average wage and salary income and the level of employment.[31] Thus, the longer-run estimates do even more to indicate that state enterprise zones did not generate beneficial effects. Still there is some state variation pointing to at least a couple instances of positive effects.

Heterogenous Effects: Types of Incentives

We next reconsider the evidence on both shorter-run and longer-run effects of state enterprise zone programs by focusing on the heterogeneity in the types of benefits offered across states. Table 6 gives a summary version of online Appendix Table A1 that shows the classification of the states in our sample based on the four features of state enterprise zones we study, including the four miscellaneous local property tax incentives we consider within the miscellaneous category.[32]

The detailed breakdown of enterprise zone benefits by state in online Appendix Table A1 indicates that there is considerable heterogeneity across states in terms of how many kinds of benefits are incorporated into the state's program. Both the number and types of benefits could be partly endogenous with respect to how disadvantaged the targeted tracts are. For example, states may be more aggressive in adopting a wider array of benefits when they are targeting relatively worse-performing areas with their enterprise zone program. Our identification strategy that matches on the pretreatment trends in outcomes between the treated and control tracts is intended to address this issue, ensuring—in this case—that we are not comparing tracts targeted by enterprise zone programs with many benefits (as in our example) to control tracts that are not performing as badly.

Table 5. Longer-Run State-Specific Estimates of the Effect of Enterprise Zones (EZs): Only Limited Controls (1990-2010).

	Unemployment rate (%)	Poverty rate (%)	Fraction of households with wage and salary income (%)	Average household wage and salary income ($2000)	Employment
EZ × California	−0.103 (0.243)	1.346 (1.399)	1.754 (1.249)	−1,404 (2,224)	−65.56 (67.49)
N (EZ)	16,505 (325)	16,505 (325)	16,505 (325)	16,505 (325)	16,505 (325)
EZ × Colorado	0.514 (2.320)	0.920 (10.56)	5.249 (7.949)	4,639*** (1,316)	−35.58 (131.3)
N (EZ)	16,194 (14)	16,194 (14)	16,194 (14)	16,194 (14)	16,194 (14)
EZ × Florida	2.554 (2.799)	−2.808 (6.553)	2.223 (3.338)	2,810 (3,878)	−50.22 (193.9)
N (EZ)	16,247 (67)	16,247 (67)	16,247 (67)	16,247 (67)	16,247 (67)
EZ × Hawaii	−2.102 (4.201)	−2.324 (9.192)	3.485 (10.23)	−13,219 (26,533)	−122.3 (362.7)
N (EZ)	16,189 (9)	16,189 (9)	16,189 (9)	16,189 (9)	16,189 (9)
EZ × Nebraska	−5.518 (11.25)	−1.635 (15.75)	12.45 (14.87)	−8,502 (14,154)	−1,229*** (279.2)
N (EZ)	16,183 (3)	16,183 (3)	16,183 (3)	16,183 (3)	16,183 (3)
EZ × New York	−2.139 (1.415)	3.602* (1.938)	2.610 (1.932)	−6,560*** (1,932)	−100.4 (76.61)
N (EZ)	16,295 (115)	16,295 (115)	16,295 (115)	16,295 (115)	16,295 (115)
EZ × Ohio	1.501*** (0.423)	2.068*** (0.466)	−1.952*** (0.540)	−8,847*** (1,232)	−202.4*** (36.09)
N (EZ)	17,161 (981)	17,161 (981)	17,161 (981)	17,161 (981)	17,161 (981)
EZ × Oregon	−2.433 (1.665)	−4.816* (2.663)	−1.228 (3.715)	−3,547 (5,375)	149.1 (118.3)
N (EZ)	16,228 (48)	16,228 (48)	16,228 (48)	16,228 (48)	16,228 (48)
EZ × Rhode Island	−1.972 (1.865)	−0.557 (2.485)	1.620 (4.206)	−14,208** (5,657)	−146.4 (181.1)
N (EZ)	16,207 (27)	16,207 (27)	16,207 (27)	16,207 (27)	16,207 (27)
EZ × Virginia	1.124 (1.969)	5.129 (3.598)	−1.584 (4.748)	−8,593** (3,852)	−358.0** (169.7)
N (EZ)	16,209 (29)	16,209 (29)	16,209 (29)	16,209 (29)	16,209 (29)

Note. Each cell is an estimate of the average longer-run (1990-2010) effect of enterprise zones from a separate propensity score matched probit model that matches treatment to control tracts using 10 covariates: 1980 and 1990 tract-level unemployment rate, poverty rate, fraction of households with wage and salary income, average wage and salary income, and employment. The estimates presented are average treatment effects on the treated. The control tracts are not restricted to the same state as the treated tracts.
Source. Authors' computations.
*$p < .1$. **$p < .05$. ***$p < .01$.

Table 6. Enterprise Zone Broad Benefit Categories by State.

	State													
Benefit	CA	CO	FL	HI	IL	MA	NE	NY	OH	OR	RI	VA	WI	Total
1. Financing	X	X	X		X			X			X			6
2. Tax credit for investments or operations	X	X	X	X	X	X	X	X	X	X	X	X	X	13
3. Miscellaneous local property tax incentives	X	X	X	X	X			X						6
4. Hiring incentives	X	X	X	X	X		X	X	X	X	X	X	X	12

Note. See online Appendix B for details. CA = California; CO = Colorado; FL = Florida; HI = Hawaii; IL = Illinois; MA = Massachusetts; NE = Nebraska; NY = New York; OH = Ohio; OR = Oregon; RI = Rhode Island; VA = Virginia; WI = Wisconsin. The "Total" column counts the number of states with hiring credits in the indicated category.
Source. Authors' coding.

We begin—for the shorter run analysis—with the state-specific, shorter-run (1990 to 2000) estimates like those in Table 3, although we use the estimates not just from Table 3 but also from corresponding estimates for the other two samples shown (not reported). As shown in Table 6, all states included in the analysis provide investment credits as part of their enterprise zone programs. Hence, the effect of investment credits is subsumed in the intercept for the models we estimate (in this case, the estimated intercept coefficient should be interpreted as the effect of enterprise zones with only this feature). In addition, when we drop Massachusetts and Wisconsin, all the remaining 10 states also include hiring

Table 7. Shorter-Run Estimates of Heterogeneous Effects of Enterprise Zone Benefits.

	Unemployment rate (%)	Poverty rate (%)	Fraction of households with wage and salary income (%)	Average household wage and salary income ($2000)	Employment
Panel 1: No limits on controls (N = 12)					
Constant (includes investment credits [12])	−1.411 (0.901)	1.236 (0.796)	−1.929 (1.080)	−2,777*** (565.8)	−92.72* (48.19)
Financing (5)	1.072 (3.617)	0.917 (3.195)	−2.658 (4.336)	2,433 (2,271)	−220.4 (193.4)
Miscellaneous local property tax incentives (5)	0.379 (3.648)	3.028 (3.222)	1.511 (4.374)	−2,319 (2,290)	240.7 (195.1)
Hiring credits (11)	−0.591 (1.110)	−2.592** (0.981)	2.235 (1.331)	1,955** (697.3)	70.79 (59.39)
Panel 2: Excluding MA and WI (N = 10)					
Constant (includes investment and hiring credits [10])	−1.490*** (0.420)	−1.912*** (0.534)	1.752*** (0.402)	−1,421* (705.7)	−10.30 (38.60)
Financing (5)	−0.886 (2.316)	4.443 (2.939)	−4.214* (2.212)	−2,163 (3,888)	−273.0 (212.6)
Miscellaneous local property tax incentives (5)	0.796 (2.335)	−0.339 (2.964)	3.655 (2.231)	2,254 (3,921)	219.3 (214.4)
Panel 3: Only limited controls (N = 10)					
Constant (includes investment and hiring credits [10])	−1.967*** (0.417)	−1.956*** (0.522)	0.489 (0.393)	−1,798** (560.1)	−33.73 (26.18)
Financing (5)	0.975 (2.296)	5.335 (2.877)	−1.244 (2.164)	−6,275* (3,086)	−232.4 (144.2)
Miscellaneous local property tax incentives (5)	0.823 (2.315)	−0.712 (2.901)	2.179 (2.182)	6,599* (3,112)	198.5 (145.4)

Note. Each column presents estimates from a separate two-stage model where the state-specific shorter-run (1990-2000) estimates in Table 3 (and corresponding estimates for the other samples) are regressed on indicators for different enterprise zone benefits. Each second-stage estimate is weighted by the number of treated tracts in that state. All states included in Panel 1 provide investment credits, so the effects of investment credits are subsumed in the constant. All states included in Panels 2 and 3 provide investment and hiring credits, so the effects of investment and hiring credits are subsumed in the constant. Number in parentheses next to each benefit category refers to the number of states that provide that benefit.
Source. Authors' computations.
$^{*}p < .1$. $^{**}p < .05$. $^{***}p < .01$.

credits, since only one state (Massachusetts) does not have hiring credits. So, when we use just the 10 states, hiring credits are also subsumed in the intercept, which unfortunately—with the cleanest sample—leaves us only with reliable, independent information on financing credits and miscellaneous local property tax incentives. Below we turn to other evidence on the effects of specific features of hiring credits.

The results for the four features of enterprise zone incentives summarized in Table 6 are reported in Table 7. The key finding is that we find relatively few significant effects of enterprise zone features, and when we do, the effects are mixed regarding whether program benefits improve labor market outcomes. For example, the effects of financing incentives, according to these results, are adverse for the fraction of households with wage and salary income (Panel 2) and average household income (Panel 3). There is a positive estimated impact of miscellaneous local property tax incentives in Panel 3 only. Panel 3 suggests beneficial effects of investment and hiring credits in terms of reducing unemployment and poverty, but also an adverse effect lowering average incomes.[33]

In Table 8, we present similar analyses, in this case using the state-specific longer-run estimates from Table 5 (and corresponding estimates for the other two samples, which, again, are not reported). There is, if anything, more consistent evidence of adverse effects, as every significant coefficient estimate is in this direction except for the estimated effect of miscellaneous local property tax incentives on employment in Panel 1.[34,35]

Heterogenous Effects: Hiring Credits

Finally, we could not learn much about whether enterprise zone programs offering hiring credits are beneficial because most states offer these benefits—and all states in our cleaner sample, excluding Massachusetts and Wisconsin, offer these benefits. However, there are differences in the nature of hiring credits across states, which we now explore and summarize in Table 9. Note that (see online Appendix Table A2) we collapse two features pertaining to where eligible workers must live to indicate whether, for the most part, credits are available only for workers who live in the enterprise zone.

Table 8. Longer-Run Estimates of Heterogeneous Effects of Enterprise Zone Benefits.

	Unemployment rate (%)	Poverty rate (%)	Fraction of households with wage and salary income (%)	Average household wage and salary income ($2000)	Employment
Panel 1: No limits on controls (N =12)					
Constant (includes investment credits [12])	0.0624 (0.687)	1.584* (0.840)	−1.420 (1.058)	−5,549** (1,680)	−146.4** (56.43)
Financing (5)	−0.398 (2.758)	−4.330 (3.371)	0.540 (4.246)	−6,393 (6,743)	−439.2* (226.5)
Miscellaneous local property tax incentives (5)	−0.565 (2.782)	3.221 (3.400)	1.144 (4.283)	10,420 (6,801)	622.2** (228.4)
Hiring credits (11)	0.333 (0.847)	0.162 (1.035)	−0.603 (1.304)	−677.8 (2,070)	−44.11 (69.55)
Panel 2: Excluding MA and WI (N =10)					
Constant (includes investment credits [10])	1.181** (0.465)	1.413 (0.834)	−0.477 (0.322)	−5,725*** (1,148)	−144.5** (59.44)
Financing (5)	−2.861 (2.559)	−0.224 (4.595)	−0.856 (1.772)	−4,461 (6,325)	−240.5 (327.5)
Miscellaneous local property tax incentives (5)	−0.137 (2.581)	−1.774 (4.634)	3.538* (1.787)	6,410 (6,378)	350.6 (330.2)
Panel 3: Only limited controls (N =10)					
Constant (includes investment credits [10])	1.260** (0.514)	1.792* (0.777)	−1.838*** (0.368)	−8,705*** (1,025)	−193.6*** (36.34)
Financing (5)	−1.964 (2.833)	−0.875 (4.279)	2.262 (2.025)	−1,335 (5,649)	48.40 (200.2)
Miscellaneous local property tax incentives (5)	0.444 (2.857)	0.305 (4.315)	1.735 (2.042)	7,987 (5,696)	74.73 (201.9)

Note. Each column presents estimates from a separate two-stage model where the state-specific longer-run (1990-2010) estimates in Table 5 (and corresponding estimates for the other samples) are regressed on indicators for different enterprise zone benefits. Each second-stage estimate is weighted by the number of treated tracts in that state. All states included in Panel 1 provide investment credits, so the effects of investment credits are subsumed in the constant. All states included in Panels 2 and 3 provide investment and hiring credits, so the effects of investment and hiring credits are subsumed in the constant. Number in parentheses next to each benefit category refers to the number of states that provide that benefit.
Source. Authors' computations.
*$p < .1$. **$p < .05$. ***$p < .01$.

Table 9. Hiring Credit Differences Across States.

	State													
Hiring credit	CA	CO	FL	HI	IL	MA	NE	NY	OH	OR	RI	VA	WI	Total
Live-in requirement to receive hiring credit		X	X				X	X			X	X		6
Credit based on wages	X		X					X			X		X	5
Flat credit per new employee		X	X		X		X	X						5
Payroll or unemployment insurance tax exemption				X					X	X		X		4

Note. A credit based on wages provides a tax credit for businesses who employ eligible employees equal to some portion (varies by state) of the total wage bill paid to those employees.
Source. Authors' coding. See online Appendix B.

The results are reported in Tables 10 and 11 for the shorter run and the longer run, respectively, using our three alternative analysis samples.[36] Paralleling our earlier findings, there is little consistent evidence that any kind of enterprise zone hiring credit delivers beneficial effects. In Table 10, for the shorter-run estimates there are three statistically significant estimates indicative of a beneficial effect (all in Panel 1)—for the effect of a flat credit per new employee on the unemployment rate, and the effects of any hiring credit on average income and employment. But there are also significant estimates in the opposite direction.

We might have expected that credits with live-in requirements would be most likely to deliver benefits to local residents, but there is no evidence of this in Table 10. There are a couple of estimates in the opposite direction (for employment in Panel 1 and the fraction with wage and salary income

Table 10. Shorter-Run Estimates of Heterogeneous Enterprise Zone Hiring Credits.

	Unemployment rate (%)	Poverty rate (%)	Fraction of households with wage and salary income (%)	Average household wage and salary income ($2000)	Employment
Panel 1. No limits on controls (N =12)					
Constant	−1.411*	1.236	−1.929*	−2,777***	−92.72**
	(0.692)	(0.958)	(0.907)	(354.0)	(30.38)
Any hiring credit (11)	0.0428	−5.932	4.538	8,521***	550.9**
	(4.324)	(5.982)	(5.666)	(2,212)	(189.8)
Live-in requirement to receive hiring credit (5)	2.595	1.024	−5.248	−126.1	−336.5**
	(2.258)	(3.124)	(2.959)	(1,155)	(99.09)
Credit based on wages (4)	1.809	6.401	−2.252	−5,512**	−424.5*
	(4.180)	(5.782)	(5.477)	(2,138)	(183.4)
Flat credit per new employee (4)	−5.800*	0.695	1.997	−2,735*	210.4
	(2.571)	(3.556)	(3.368)	(1,315)	(112.8)
Payroll or unemployment insurance tax exemption (4)	−0.723	3.391	−2.098	−6,584**	−469.4**
	(4.296)	(5.943)	(5.630)	(2,197)	(188.5)
Panel 2. Excluding MA and WI (N =10)					
Constant (includes any hiring credit (10))	−1.777	−1.187	2.903	1,976	408.8
	(3.454)	(3.738)	(1.760)	(4,497)	(249.5)
Live-in requirement to receive hiring credit (5)	2.448	1.953	−4.060***	−4,735	−223.7
	(1.827)	(1.977)	(0.931)	(2,379)	(131.9)
Credit based on wages (4)	0.313	3.955	−0.829	−1,966	−410.3
	(3.377)	(3.655)	(1.721)	(4,398)	(243.9)
Flat credit per new employee (4)	−3.250	−3.334	1.600	1,205	18.90
	(2.079)	(2.250)	(1.059)	(2,707)	(150.2)
Payroll or unemployment insurance tax exemption (4)	0.241	−0.809	−1.009	−3,251	−411.6
	(3.476)	(3.762)	(1.771)	(4,527)	(251.1)
Panel 3. Only limited controls (N =10)					
Constant (includes any hiring credit (10))	−3.056	−4.191	2.148	6,016	123.9
	(3.878)	(3.676)	(3.481)	(5,137)	(194.1)
Live-in requirement to receive hiring credit (5)	1.543	2.377	−2.020	−4,359	−143.3
	(2.051)	(1.944)	(1.841)	(2,717)	(102.7)
Credit based on wages (4)	2.698	6.943	−0.227	−7,122	−145.4
	(3.792)	(3.595)	(3.404)	(5,023)	(189.8)
Flat credit per new employee (4)	−1.175	−2.313	0.612	2,565	−10.18
	(2.335)	(2.213)	(2.096)	(3,093)	(116.9)
Payroll or unemployment insurance tax exemption (4)	1.062	2.165	−1.591	−7,665	−150.9
	(3.903)	(3.700)	(3.504)	(5,171)	(195.4)

Note. Each column presents estimates from a separate two-stage model where the state-specific shorter-run (1990-2000) estimates in Table 3 (first stage), and corresponding estimates for other samples are regressed on indicators for different enterprise zone benefits (outlined in Table 9 and described in more detail in the online Appendix Tables A1 and A2). Each second-stage estimate is weighted by the number of treated tracts in that state. The overall effect of hiring credits is not estimable in Panels 2 and 3 because Massachusetts was the only state to not provide hiring credits, and Massachusetts is dropped from these analyses.
Source. Authors' computations.
*$p < .1$. **$p < .05$. ***$p < .01$.

in Panel 2). In the longer-run estimates, in Table 11, there is again mixed evidence in general, and, for live-in requirements, only evidence of an adverse effect (on employment, in Panel 1).

Overall, when we look at program heterogeneity, the estimates are often much less precise, which is not surprising since the evidence is based on subsets of states. For example, comparing the uniform shorter-run effects in Table 2 with the heterogeneous effects in Table 7, the standard errors on specific program features are larger. Moreover, the estimates are sometimes remarkably imprecise (see, e.g., the estimated effects of miscellaneous credits on all outcomes, and of hiring credits in Panel 1). The same is generally true of the longer-run effects. These findings suggest that it is difficult,

Table 11. Longer-Run Estimates of Heterogeneous Enterprise Zone Hiring Credits.

	Unemployment rate (%)	Poverty rate (%)	Fraction of households with wage and salary income (%)	Average household wage and salary income ($2000)	Employment
Panel 1. No limits on controls (N = 12)					
Constant	0.0624	1.584*	−1.420	−5,549**	−146.4**
	(0.599)	(0.722)	(1.164)	(1,954)	(57.46)
Any hiring credit (11)	1.096	−9.369*	4.701	14,717	375.1
	(3.742)	(4.513)	(7.275)	(12,208)	(358.9)
Live-in requirement to receive hiring credit (5)	−0.417	−2.868	−2.225	−5,518	−497.9**
	(1.954)	(2.357)	(3.799)	(6,375)	(187.4)
Credit based on wages (4)	−0.729	7.974	−3.603	−11,072	−287.3
	(3.617)	(4.363)	(7.032)	(11,801)	(347.0)
Flat credit per new employee (4)	−2.327	4.571	1.947	2,479	580.3**
	(2.224)	(2.683)	(4.325)	(7,257)	(213.4)
Payroll or unemployment insurance tax exemption (4)	−0.745	9.707*	−5.266	−15,156	−400.7
	(3.718)	(4.484)	(7.228)	(12,129)	(356.6)
Panel 2. Excluding MA and WI (N = 10)					
Constant (includes any hiring credit (10))	−0.702	−4.401	0.566	2,679	377.2
	(3.218)	(6.536)	(3.522)	(10,252)	(593.1)
Live-in requirement to receive hiring credit (5)	−0.615	0.0613	−1.907	−3,660	−146.8
	(1.702)	(3.457)	(1.863)	(5,423)	(313.7)
Credit based on wages (4)	−0.289	5.189	1.409	−4,915	−387.3
	(3.146)	(6.391)	(3.444)	(10,025)	(579.9)
Flat credit per new employee (4)	−1.716	−3.233	2.441	−1,289	26.36
	(1.937)	(3.935)	(2.120)	(6,172)	(357.0)
Payroll or unemployment insurance tax exemption (4)	1.930	5.812	−0.969	−8,275	−516.4
	(3.238)	(6.578)	(3.545)	(10,319)	(596.9)
Panel 3. Only limited controls (N = 10)					
Constant (includes any hiring credit (10))	−0.312	0.356	5.790*	3,503	−226.2
	(5.020)	(7.734)	(2.381)	(10,247)	(356.2)
Live-in requirement to receive hiring credit (5)	−0.964	0.921	0.0895	−5,982	−129.6
	(2.655)	(4.091)	(1.259)	(5,420)	(188.4)
Credit based on wages (4)	0.139	0.773	−4.053	−5,430	164.4
	(4.909)	(7.563)	(2.328)	(10,020)	(348.3)
Flat credit per new employee (4)	0.725	−0.809	0.641	4,798	109.5
	(3.022)	(4.656)	(1.433)	(6,169)	(214.4)
Payroll or unemployment insurance tax exemption (4)	1.621	1.423	−7.655**	−11,979	39.60
	(5.053)	(7.784)	(2.396)	(10,313)	(358.5)

Note. Each column presents estimates from a separate two-stage model where the state-specific longer-run (1990-2010) estimates in Table 5 (first stage), and corresponding estimates for other samples, are regressed on indicators for different enterprise zone benefits (outlined in Table 9 and described in more detail in the online Appendix Tables A1 and A2). Each second-stage estimate is weighted by the number of treated tracts in that state. The overall effect of hiring credits is not estimable in Panels 2 and 3 because Massachusetts was the only state to not provide hiring credits, and Massachusetts is dropped from these analyses.
Source. Authors' computations.
*$p < .1$. **$p < .05$. ***$p < .01$.

at least in the data we used, to pin down the effects of different enterprise zone program features.

Conclusions

Our analysis in this article sought to explore two questions regarding the effectiveness of state enterprise zones: Is there stronger evidence of beneficial effects of enterprise zones in the longer run, when the programs may have more chance to jump-start economic activity? Also, is there evidence that some particular state enterprise zone features deliver some of the intended benefits of these programs, despite the absence of overall beneficial effects of state enterprise zone programs? Unfortunately, neither looking at state enterprise zone programs through a longer-term lens nor through the lens of program heterogeneity leads to much if any consistent indication of beneficial effects of state enterprise zone programs.

Overall, we think this evidence should be interpreted cautiously. The data provide limited capacity to test for program heterogeneity, as there are not that many state programs to study and hence not a great deal of program variation. In addition, there could be richer dynamics associated with longer-run effects that our decadal analysis with census data does not capture. Moreover, there could be other sources of variation that are relevant, such as unmeasured initial economic (or other) conditions in the areas designated as enterprise zones.

Nonetheless, our evidence, as it stands, buttresses the generally negative assessments of the effects of enterprise zones. One reaction is to view this article as one more "nail in the coffin" of enterprise zones, and to conclude that we need to look for alternative place-based policies, perhaps at a more aggregate labor market level (as argued by Bartik, 2020), or simply rely on people-based policies that offer benefits and credits based on individual or family characteristics rather than location characteristics. However, Neumark (2018) also argued that it may be possible to draw on lessons from research on enterprise zones, hiring credits, alternative policies, and labor market networks, to develop more effective enterprise zones. The key features of these would include aggressive job subsidies, building skills for private sector employment, revitalization of targeted areas, local hiring, and partnerships with local nonprofits, employers, and community groups. If one believes we still need to try to target policies to local economic development at the neighborhood level, then there is a strong case for both abandoning enterprise zones as we have done them in the past and considering radically different alternatives that could be more effective.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: The research for this article was funded by Laura and John Arnold Foundation.

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Supplemental Material

Supplemental material for this article is available online.

Notes

1. Neumark and Young (2019) cast serious doubt on the magnitudes of the effects Ham et al. estimated, in part because of data errors and in part because of estimation strategies—in particular, the control tracts used. We do not rehash that material here but suffice it to say that we find very little evidence of positive effects of either state or federal enterprise zones. That article also discussed the differences between Empowerment Zones and Enterprise Communities; these federal programs are not the focus of our article.
2. Estimated, if not actual effects, could also diminish over time because of greater measurement error (contamination) from changes in enterprise zone boundaries, variation in incentives, and such. We have tried as best we can to account for these and we did not find evidence of changes in incentives for the period we studied.
3. See Table 18.2 in Neumark and Simpson (2015) for an exhaustive summary of the more recent research on enterprise zones, and other overviews covering earlier research in Elvery (2009), Ham et al. (2011), and Wilder and Rubin (1996).
4. An early version of Ham et al. (2011), written by Swenson (2008), provided an overview of various state enterprise zone programs in an appendix.
5. Some examples include California tax form 3805z, which is used by businesses claiming enterprise zone tax incentives and is available from 1994 to 2015; the Colorado Department of Revenue provides form FYI Income 22 to businesses claiming enterprise zone tax incentives; and Massachusetts provides Schedule EOA 1099 form to businesses claiming economic opportunity area credits.
6. For example, Hawaii's Department of Business produced a report entitled "Hawaii's Enterprise Zones Partnership Program" in 2007.
7. For example, Dowall (1996).
8. For reasons explained below, Illinois gets dropped from our analysis. We nonetheless retain it in the description of policy variation, in case a reader wants to examine the Ham et al. (2011) state-specific estimates in light of this program variation. (Illinois remains in their data because they do not use the Neighborhood Change Database [NCDB]; see Neumark & Young, 2019).
9. As documented below, many of the others seem quite minor, and nearly all states offer some benefit we otherwise grouped in this miscellaneous category.
10. One such attempt in relation to hiring credits is described below.
11. Their data can be downloaded from https://sites.google.com/usc.edu/ayse-imrohoroglu/programs?authuser=0.
12. Our goal was to avoid contaminated controls in the form of tracts designated as enterprise zones in the 2000s. Some tracts lost their designation in the 2000s, although it is difficult to pin down the timing, as explained in online Appendix C. However, given that we are estimating the longer-run effects of zones established in the 1990s, the ending of a tract's designation as an enterprise zone in the 2000s is not problematic, although clearly it could be of interest to distinguish between tracts designated in the 1990s that remained enterprise zones in the 2000s and those that no longer did.

13. We requested this information from the authors and they said they did not have it and hence could not provide it. (They only offered the name of a consultant we could pay to, apparently, recover the zone maps they used, but even then it was not clear the source material was available.)
14. We attempted to gather this information from all of the states in our NCDB sample, but historical data were no longer available for other states.
15. This made the biggest difference for Ohio, where we found hundreds of additional tracts designated as enterprise zones. For California there were an additional 140 tracts. Still the results were qualitatively similar using Ham et al.'s coding (results available on request from the corresponding author), which would not be surprising if—as most of our evidence suggests—the true effects of enterprise zones are close to zero.
16. For some states where we determine enterprise zone designation in the 2000s or, in the case of the three states for which we determine enterprise zone designation in the 1990s, we have to draw enterprise zone boundaries in ArcGIS by referencing physical maps. Because this is done by hand, there are some cases where we may unintentionally overlap an enterprise zone boundary with the boundary of a tract that does not have an enterprise zone. Therefore, we code tracts with less than 1% of overlap with an enterprise zone as controls to account for this error.
17. The perception of researchers that the NCDB matches tracts well is reflected in the extensive—indeed, nearly exclusive—use of the NCDB data in research that matches tracts over time to study enterprise zones, other local programs, and different questions entirely. See, for example, Oakley and Tsao (2006, 2007) and Meltzer (2012) on enterprise zones; and Cameron and McConnaha (2006), Card et al. (2008), Easterly (2009), and Logan and Zhang (2010) on a range of other topics.
18. See https://s4.ad.brown.edu/projects/diversity/researcher/LTDB2.htm.
19. They also implement an estimator using all potential control tracts in the state that are also not designated as federal zones, in a somewhat different framework. However, this is less comparable to our propensity score strategy so we do not discuss it further. Also, it has the same problems as the other controls Ham et al. use (Neumark & Young, 2019).
20. Another potential problem with using nearby or contiguous control tracts is that there may be negative spillovers that, if present, create a bias toward finding positive effects in treated tracts. Hanson and Rohlin (2013) found evidence of this effect, and hence bias, in estimating the effects of federal Empowerment Zones on employment and on establishment counts.
21. An alternative approach to estimating the effects of enterprise zones, applicable to some settings, is to exploit discontinuities in area characteristics (such as tract poverty rates) that make areas eligible or ineligible (e.g., Billings, 2009; Freedman, 2012, 2013).
22. We do not restrict potential control tracts to be in the same state as the treated tract, as tracts in other states can potentially provide better controls. This is not that unusual in evaluating the effects of local policies. For example, in a recent paper, Allegretto et al. (2018) estimated the effects of city minimum wages by matching on control counties in other states. The rationale for this approach is that other urban areas in other states may provide better counterfactuals because their experiences and conditions are similar, and the differences between treatment and control tracts within a state could more likely reflect the endogenous selection of treatment tracts. That said, matching approaches are inherently backward-looking; one potential criticism of using controls from other states is that effects could be confounded by other state policies adopted posttreatment. We are not aware of other significant policies affecting disadvantaged urban areas that changed in our sample period, and indeed other—nonpolicy—shocks may be more common to these areas across states. In addition, when we attempted to find controls only within states, we often failed to find good matches.
23. Matching on the 1980 to 1990 change and the level for 1 year would yield identical results. We reasoned that matching on levels, and not only changes, would provide better controls than just matching on changes. This is the procedure used in Neumark and Young (2019), as well as earlier in Bondonio and Engberg (2000).
24. Propensity score matching was performed using the "teffects psmatch" command in Stata 13. Controls used for matching were the 1980 levels and 1990 levels for the unemployment rate, poverty rate, average wage and salary income, fraction with wage and salary income, and count of employed persons (i.e., our outcomes). The propensity score is estimated using a probit model and we selected the option "atet" to estimate the average treatment effect on the treated.
25. Interestingly, there is virtually no attention to pretrends in the U.S. enterprise zone literature. O'Keefe (2004), Greenbaum and Engberg (2004), Elvery (2009), and Busso et al. (2013) all matched on levels at a single point in time (as did, e.g., Gobillon et al. [2012] in research on French enterprise zones). We have only found one study by Bondonio and Engberg (2000) that studied effects of enterprise zones on employment growth matching on employment growth in prior periods, paralleling what we do here. (They find a negative but insignificant effect of past growth on zone designation.)
26. See the discussion of the estimator at https://www.ssc.wisc.edu/sscc/pubs/stata_psmatch.htm, which is based on Abadie and Imbens (2016).
27. The results were similar for the samples corresponding to Panels 1 and 2 of Table 2. (Results available on request from the corresponding author.)
28. Note that for Nebraska there is a large negative, but very imprecise, estimated effect on the unemployment rate. However, there are only three enterprise zone tracts in Nebraska.
29. We assume that if we do not have data showing a tract continuing to be designated as an enterprise zone in the 2000s, then it was de-designated. Given our uncertainty about developments in Massachusetts and Wisconsin, this analysis is best done excluding these two states. We also restrict the analysis to the control tracts considered in Panel 3 because, as argued above, these provide our cleanest evidence.
30. The results were similar for the samples corresponding to the other panels in Table 4. (Results available on request from the corresponding author.)

31. The negative effect on average wage and salary income is never mirrored in a positive effect on the fraction with wage and salary income, which if it occurred could indicate lower earners entering the labor market.
32. There was no stronger evidence of enterprise zone effectiveness when we included all the miscellaneous benefits in a broader category. (Results available on request from the corresponding author.)
33. Online Appendix Table D2 reports estimates for the same specifications and samples, but unweighted. Note that some coefficients are identified for only one state, in which case the weighting does not affect the estimated coefficient. Note also that some estimates are much more precise, owing to the weighting substantially reducing the influence of some states. Overall, the evidence is again in contradictory directions, and without weighting some of the estimates seem implausibly large (in both directions). Thus, we read this analysis, too, as failing to provide consistent evidence of beneficial effects of particular enterprise zone features.
34. Online Appendix Table D3 reports estimates for the same specifications and samples, but unweighted. The same comments regarding online Appendix Table D2 apply. In this table, evidence on investment credits is generally much weaker, although the adverse effects still persist for two outcomes in Panel 3. There is evidence of significant beneficial effects of miscellaneous local property tax incentives in Panel 1, but these disappear in Panels 2 and 3. Again, there is no consistent evidence of beneficial effects of particular enterprise zone benefits.
35. With respect to the earlier discussion of the intensity of program benefits, we did try to classify the intensity of the key enterprise zone benefit—hiring credits—as weak or strong. We then re-ran our models for program heterogeneity (Tables 7 and 8) breaking the hiring credit variable into these two policies. These reestimations did not provide any additional evidence that hiring credits help. In particular, there is no evidence of positive impacts of strong hiring credits. See online Appendix E.
36. The results for the estimates without weighting are reported in online Appendix Tables D4 and D5. The findings are qualitatively similar.

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